

RRC-Raptor-V4H

CANFD

- User's Manual: Software

V0.0.3

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i. VERSION LIST

Description	Version	Date	Author
RTX_CANFD	V0.0.1	2024/07/23	Jerry
RTX_CANFD	V0.0.2	2024/10/29	Jerry
Translate the content into English.	V0.0.3	2024/11/20	Jerry

ii. HASHTAG

#V4H

#CAN

#CANFD

#BSP

1. Overview

1.1. Overview

This manual explains the driver module (this module) that controls the CANFD (RS-CANFD) on R-Car

1.2. Function

The RS-CANFD controller has the following two interface modes.

- Classical CAN mode: Only classical CAN frames are handled.
- CAN FD mode: both the classical CAN frames and CAN FD frames are handled.

Two modes use different register maps with the same base address, and the register maps are switched by switching the modes.

This module puts the controller in CANFD mode by default. To switch the controller to Classical CAN mode, add "renesas,no-can-fd" optional property to the device tree node. The RS-CANFD controller reset is required to switch modes.

2. Directory Configuration

The directory configuration is shown below.

—— `drivers/net/can/rcar/rcar_canfd.c`

Figure 1. Directory Configuration

3. Kernel configuration

To enable the function of this module, make the following setting with Kernel Configuration. The changes affect after re-building the kernel.

```
[*] Networking support --->
    <*> CAN bus subsystem support --->
        ...
        CAN Device Drivers --->
            <*> Renesas R-Car CAN FD controller
```

Figure 2 Kernel configuration

4. Switch For Interface Mode

This module puts the controller in CANFD mode by default. To switch the controller to Classical CAN mode, add "renesas,no-can-fd" optional property to the device tree node. The RS-CANFD controller reset is required to switch modes.

```
&canfd {  
    pinctrl-0 = <&canfd0_pins>;  
    pinctrl-names = "default";  
+    renesas,no-can-fd;  
    status = "okay";  
  
    channel0 {  
        status = "okay";  
    };  
};
```


5. Connect

Provide a CAN interface so that users can add their own CAN devices.

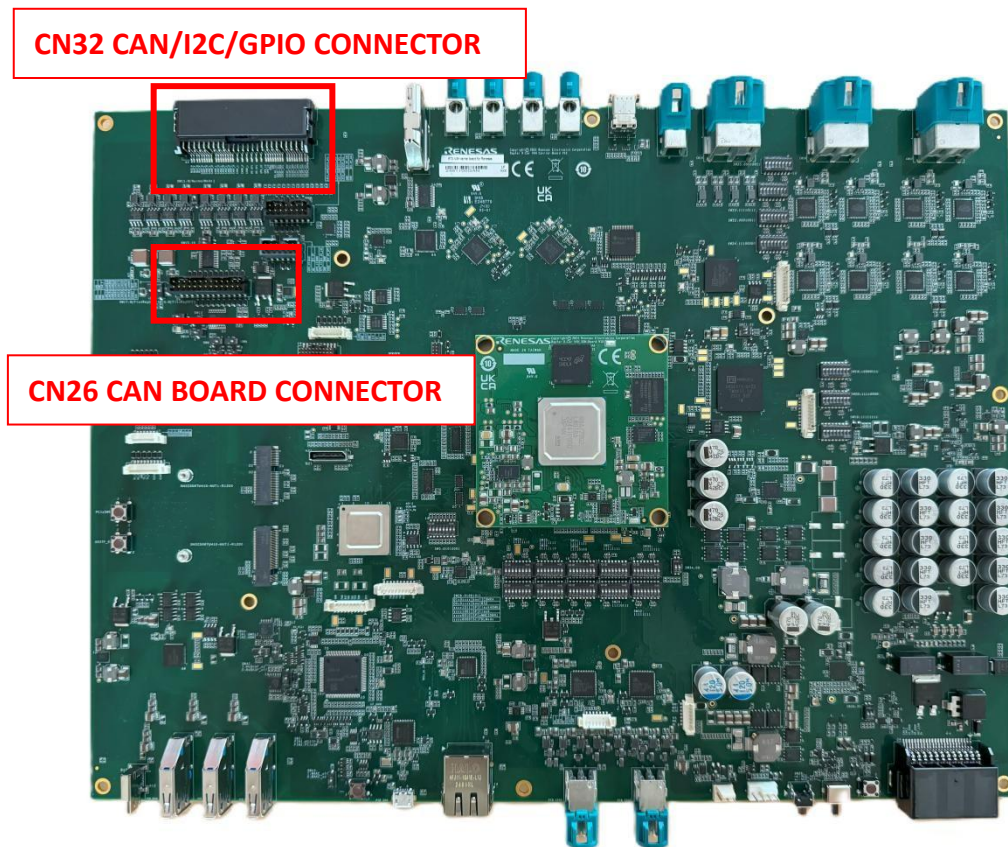


Figure 3 Connect

CN32 : CN32 CAN/I2C/GPIO CONNECTOR

CAN H / CAN L signals after passing through the internal transceiver.

PIN	SYMBOL
11	D3.3V_I2C
12	SG
19	D5.0V1
20	D5.0V1
21	CANL0_CN
22	CANH0_CN
23	CANL1_CN
24	CANH1_CN
25	CANL2_CN
26	CANH2_CN
27	CANL3_CN
28	CANH3_CN
29	CANL4_CN
30	CANH4_CN
31	CANL5_CN
32	CANH5_CN
33	CANL6_CN
34	CANH6_CN
35	CANL7_CN
36	CANH7_CN

CN26 : CAN BOARD CONNECTOR

CANFD_RX / CANFD_TX signals directly output from the SoC.

PIN	SYMBOL
1	D12.0V2
2	D5.0V1
3	CANFD0_RX_D
4	CANFD4_RX_D
5	CANFD0_TX_D
6	CANFD4_TX_D
7	CANFD0_TX_D
8	CANFD4_TX_D
9	CANFD1_RX_D
10	CANFD5_RX_D

11	CANFD1_TX_D
12	CANFD5_TX_D
13	CANFD2_RX_D
14	CANFD6_RX_D
15	CANFD2_TX_D
16	CANFD6_TX_D
17	CANFD3_RX_D
18	CANFD7_RX_D
19	CANFD3_TX_D
20	CANFD7_RX_D
21	CAN_SILENT
22	CANFD7_TX_D
23	GROUND
24	GROUND
25	GROUND
26	GROUND

6. DIP SWITCH

Switch between CANFD1, CANFD5, FlexRayA, or FlexRayB.

SW7	SW8	SW9	SW10	SW12	SW13	SW14	SW15	Select
2- 1	2- 3	2- 1	2- 1	10	2- 1	2- 1	10	CANFD1 CANFD5
2- 3	2- 1	2- 3	2- 3	01	2- 3	2- 3	01	FlexRay A,B

Refer to the dip switch settings in the 103_RRC_Raptor-V4H_Switch document.

Switch to CAN_SILENT Mode

SW11	Mode
1	Normal Mode
0	Silent Mode

Refer to the dip switch settings in the 103_RRC_Raptor-V4H_Switch document.

7. Can-utils Tool

Use can-utils can read / write can device.

Here is an example of Raptor connecting of CAN0 and CAN1.

Example Environment:

The following describes how to set up the environment using either CN32 or CN26.

- i. When using CN32, connect CANL0_CN to CANL1_CN, and CANH0_CN to CANH1_CN.
- ii. When using CN26, an external CAN transceiver is required to convert the signals CANFD0_RX_D / CANFD0_TX_D and CANFD1_RX_D / CANFD1_TX_D to CANL0_CN / CANH0_CN and CANL1_CN / CANH1_CN. Then, connect CANL0_CN to CANL1_CN and CANH0_CN to CANH1_CN.

Usage of the Tool

1. Initialize can0/can1, set to CAN FD mode, with a bitrate of 50K and a data bitrate of 1M.

```
ip link set can0 down type can
ip link set can0 type can bitrate 50000 dbitrate 1000000 fd on
ip link set can0 up type can

ip link set can1 down type can
ip link set can1 type can bitrate 50000 dbitrate 1000000 fd on
ip link set can1 up type can
```

2. Use ifconfig to verify if can0/can1 is properly up.

```
can0      Link encap:UNSPEC HWaddr 00-00-00-00-00-00-00-00-00-00-00-00-00-00-00-00
UP RUNNING NOARP MTU:72 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:10
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

can1      Link encap:UNSPEC HWaddr 00-00-00-00-00-00-00-00-00-00-00-00-00-00-00-00
UP RUNNING NOARP MTU:72 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:10
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)
```

3. Send data from can1 (using cansend) to can0 (received by candump).

```
root@v3x:~# candump can0&  
[1] 320  
root@v3x:~# cansend can0 300##0AC.AB.AD.AE.75.49.AD.D1.12.34.56.78.90.ab.cd.ef  
root@v3x:~# can0 300 [16] AC AB AD AE 75 49 AD D1 12 34 56 78 90 AB CD EF
```

4. Data sent from can1 can be received on can0.

8. APPENDIX